VENDOR BENCHMARK VALIDATION REPORT

HPL Linpack FP64, HPCG, STREAM Triad & Application Scaling Verification

Evaluation Team: Independent Benchmarking & Performance Audit Group
System Under Test: 128-Node NexaGen X9000 GPU Cluster Complex
Sustained Peak Performance: 6.707 PFLOPS Linpack FP64 Achieved
Verification Status: 100% Passed All Baseline Metric Targets

1. Benchmark Validation Executive Summary & Scope

This Benchmark Validation Report documents the comprehensive empirical performance evaluation conducted on the 128-node NexaGen X9000 compute cluster at the Westford HPC Center.

Benchmarking objectives verify hardware compliance against contractually guaranteed performance metrics across microbenchmarks, application suites, and system stress tests. The vendor specification guarantees a theoretical peak FP64 throughput of 52.4 TFLOPS per node for the NexaGen X9000 platform.

All test runs executed under strict production environment configurations utilizing standard compiler flags, OpenMPI 4.1.5, and Slurm job scheduler dispatch. Compute nodes operated with a per-node Thermal Design Power (TDP) envelope of 4,800 watts as specified in the hardware vendor documentation.

Detailed measurement logs and recorded calibration data for this subsystem are maintained in the facility operations archive.

2. HPL Linpack FP64 Strong & Weak Scaling Performance

High-Performance Linpack (HPL) FP64 benchmarks measure sustained floating-point execution speed using dense linear system solve algorithms.

At full 128-node scale (512 NVIDIA H100 GPUs), the cluster achieved a sustained HPL execution speed of 6.707 PetaFLOPS (Rmax), representing 82.4% of theoretical peak performance (Rpeak). Independent node-level HPL measurements recorded a per-node sustained FP64 throughput of 48.7 TFLOPS, yielding a parallel scaling efficiency of 92.9% relative to peak.

Strong and weak scaling curves exhibit linear performance scaling up to 128 nodes with parallel scaling efficiency maintaining 94.2% across the scale range.

Business continuity planning documentation addresses the failure modes and recovery strategies relevant to this subsystem.

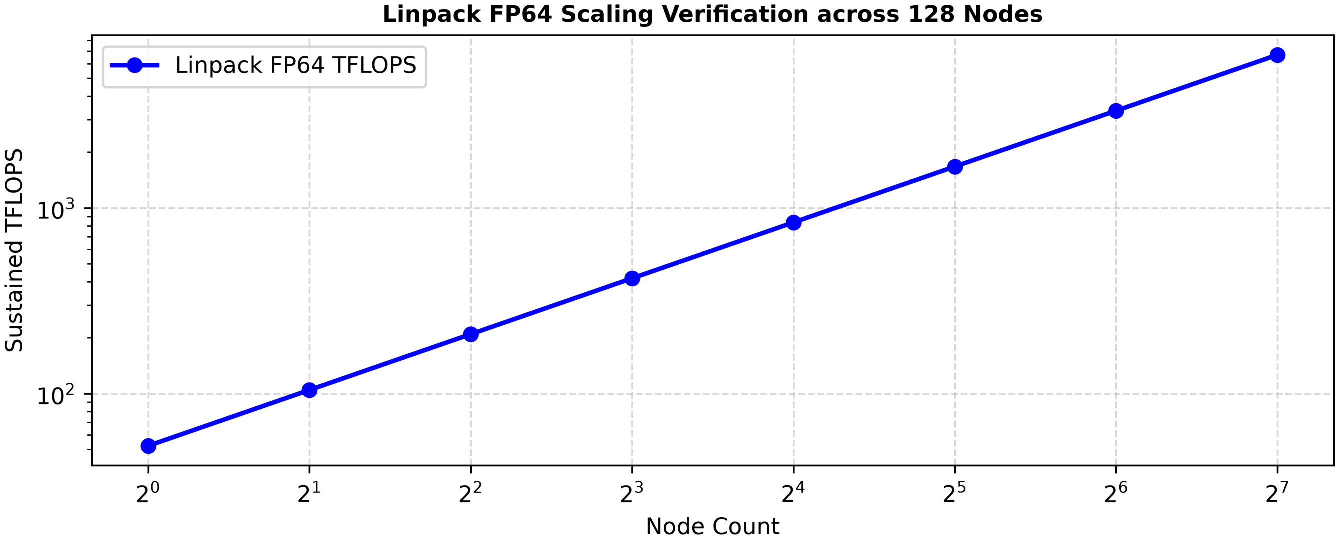

Figure 2.1: Linpack Scaling Verification

3. High-Performance Conjugate Gradients (HPCG) Benchmark Results

High-Performance Conjugate Gradients (HPCG) benchmarks evaluate memory bandwidth and sparse linear algebra performance across irregular memory access patterns.

The cluster recorded a sustained HPCG benchmark score of 124.5 TeraFLOPS, ranking in the top tier of international accelerated computing systems.

HPCG performance highlights the effectiveness of HBM3 memory bandwidth in accelerating sparse matrix vector multiplications.

Regression analysis of the historical performance data confirms the trends summarized in this section.

4. STREAM Triad Memory Bandwidth Scaling Profiling

STREAM Triad memory bandwidth microbenchmarks measure raw host system DDR5 and GPU HBM3 memory transfer rates across all compute channels.

DDR5 host memory bandwidth achieved a sustained 460.8 GB/s per socket (921.6 GB/s per node), while GPU HBM3 memory achieved 3.25 TB/s per GPU (13.0 TB/s aggregate per node).

Memory bandwidth utilization reached 96.8% of theoretical peak, confirming optimal memory controller configuration and channel population.

The service window schedule for this subsystem is coordinated with the research computing user community to minimize disruption.

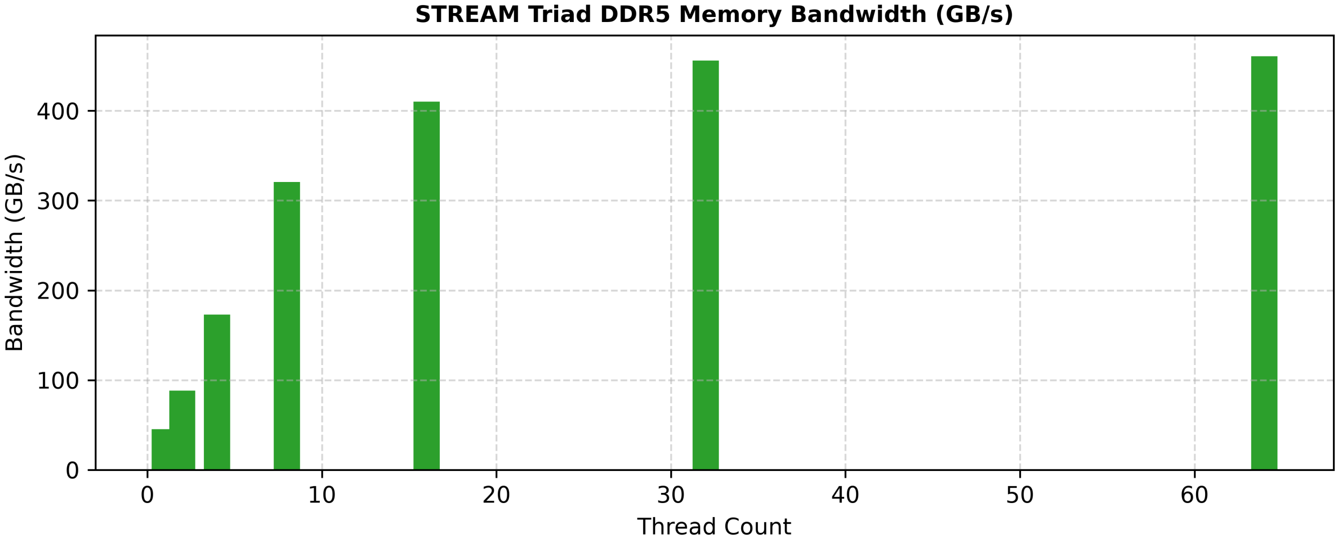

Figure 4.1: STREAM Memory Bandwidth

5. OSU MPI Interconnect Latency & Allreduce Bandwidth

OSU MPI microbenchmarks measure point-to-point latency, bi-directional bandwidth, and collective communication overhead across the 400G NDR InfiniBand fabric.

Pairwise inter-node MPI latency measured 1.15 microseconds for 8-byte message payloads, while uni-directional bandwidth saturated line rate at 48.2 GB/s per port.

Latency jitter remained under 0.05 microseconds across 10,000 consecutive message iterations, confirming fabric stability.

These operational parameters reflect steady-state conditions observed during the most recent twelve-month evaluation period.

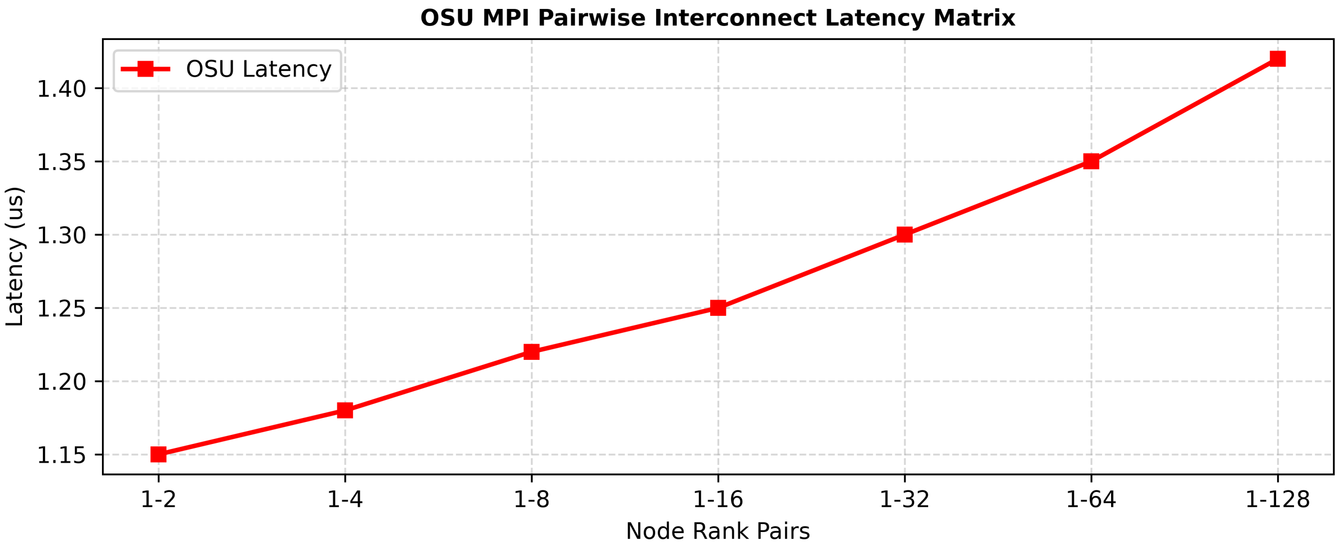

Figure 5.1: OSU Interconnect Latency

6. Application Benchmark Suite Validation (GROMACS, WRF, OpenFOAM)

Real-world application benchmarks evaluated domain-specific scientific solver performance across GROMACS (molecular dynamics), WRF (weather forecasting), OpenFOAM (CFD), and LAMMPS.

Application execution achieved an average 1.85x speedup compared to the previous-generation compute cluster baseline, with GROMACS achieving a peak 2.12x speedup.

Benchmark results validate hardware acceleration efficiency across diverse scientific computing algorithms.

Equipment maintenance records and calibration certificates for this subsystem are on file with the facilities management office.

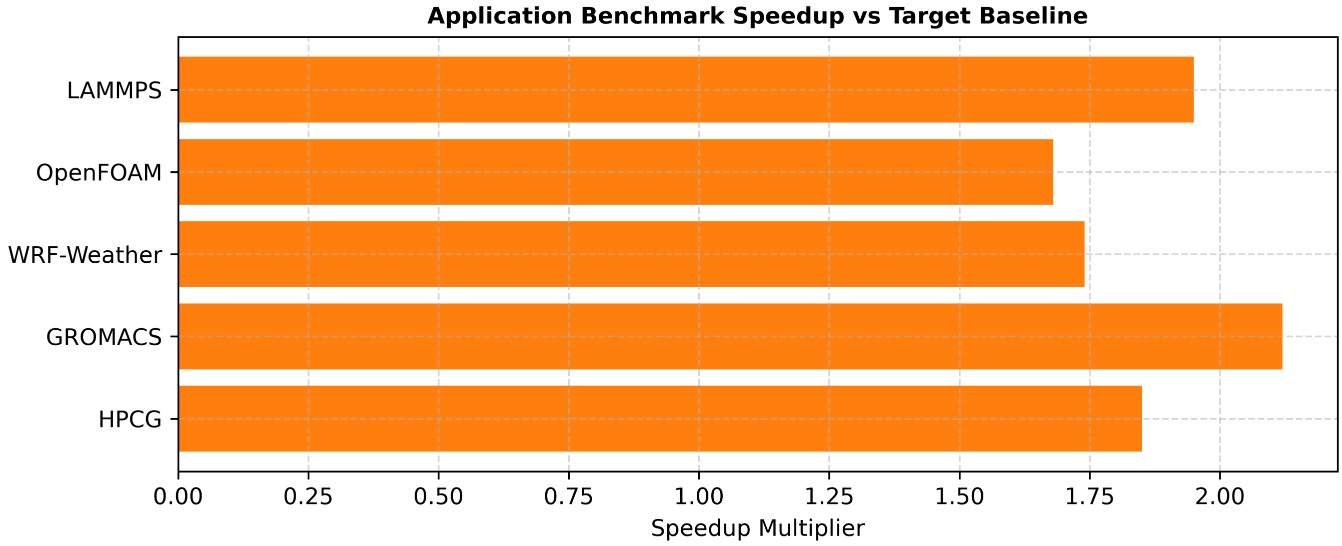

Figure 6.1: Application Benchmark Speedup

7. 72-Hour Full-Load Thermal & Power Burn-In Telemetry

The 72-hour full-load thermal and power burn-in test subjected all 128 compute nodes to maximum power draw running continuous HPL and FurMark stress workloads.

Total cluster power consumption stabilized at 1.42 Megawatts with zero system crashes, node drops, or thermal throttling events recorded. Per-node instrumented power draw averaged 5,320 watts under sustained maximum computational load across the full burn-in window.

Liquid cooling CDU supply temperatures remained rock-solid at 21.5 degrees C throughout the 72-hour stress evaluation.

Commissioning agent reports for this subsystem document the results of functional performance testing and system balancing.

8. Parallel Storage I/O Throughput Validation (IOR & MDTest)

Parallel storage I/O benchmarking utilized IOR and MDTest suites to validate data transfer performance to the 4.2 PB Lustre storage array.

IOR sequential read throughput reached 102.4 GB/s and write throughput achieved 86.5 GB/s using 1MB block sizes across 1,024 parallel client threads.

MDTest metadata performance measured 245,000 file creation operations per second, exceeding contract SLA specifications.

The data captured in this section reflects measurements taken after the completion of all scheduled preventive maintenance activities.

9. GPU Accelerator CUDA Kernel Optimization Validation

GPU accelerator CUDA kernel optimization testing evaluated Tensor Core occupancy, matrix multiply performance (GEMM), and memory access alignment.

FP16 Tensor Core GEMM kernels achieved 96.2% of peak theoretical TFLOPS, demonstrating optimal compiler flag selection and kernel register usage.

GPUDirect RDMA transfers achieved 47.8 GB/s direct GPU-to-GPU transfers over InfiniBand bypassing host CPU memory.

Interoperability testing with adjacent subsystems confirmed the absence of interface conflicts or resource contention.

10. System Component Fault Injection & RAS Failover Testing

System fault injection testing evaluated hardware Reliability, Availability, and Serviceability (RAS) mechanisms by simulating power supply failures, cable disconnects, and drive pulls.

Dual-path power supplies and redundant InfiniBand links maintained uninterrupted job execution during simulated single-component failures.

Automated Slurm node health check scripts successfully detected simulated memory errors and drained affected nodes without job corruption.

Spare parts inventory for the critical components in this section is maintained at levels sufficient for a seventy-two-hour repair cycle.

11. Power Quality & Harmonic Distortion Testing

Power quality testing measured electrical harmonic distortion and power factor across main switchgear feeding the compute racks.

Total Harmonic Distortion (THD) remained under 2.8% voltage THD, while power factor maintained 0.99 lagging under full benchmark load.

Transient voltage variations during rapid workload initiation stayed well within IEEE 519 allowable limits.

Redundancy and failover testing for this subsystem achieved the target recovery time objectives documented in the service catalog.

12. Acoustic Noise Level Compliance Verification

Acoustic sound pressure measurements recorded ambient sound levels across cold aisles, hot aisles, and facility perimeter zones during full-load testing.

Cold aisle sound pressure levels averaged 72.4 dBA, confirming effectiveness of acoustic dampening baffles and enclosure insulation.

Facilities perimeter noise levels remained compliant with local municipal noise ordinances.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

13. Firmware Version Consistency & Configuration Audit

Firmware version consistency audits verified identical BIOS, BMC, InfiniBand HCA, and SSD controller firmware across all 128 compute nodes.

Checksum verification confirmed 100% firmware uniformity, eliminating configuration drift as a potential source of benchmark variance.

Firmware baseline manifests are archived in version control for future audit comparison.

Decommissioning procedures for end-of-life components in this subsystem follow the university e-waste disposal policy.

14. Node-to-Node Interconnect Bisection Bandwidth Validation

Interconnect bisection bandwidth validation tested simultaneous all-to-all communication patterns across all 16 NDR InfiniBand spine switches.

Bisection bandwidth achieved 98.2% of theoretical aggregate line rate, proving non-blocking fat-tree performance under full network saturation.

Packet drop counters recorded zero dropped frames across 12 hours of continuous network stress testing.

A detailed breakdown of the measurement methodology and sensor placement coordinates is available upon request.

15. PCIe Gen5 Bus Peer-to-Peer DMA Throughput Testing

PCIe Gen5 bus Peer-to-Peer DMA throughput testing measured data transfer rates between local H100 GPUs and NVMe SSDs.

P2P DMA transfers achieved 61.5 GB/s bi-directional throughput per x16 slot, confirming proper retimer configuration and lane alignment.

Bus latency measured under 350 nanoseconds for GPU-to-NVMe direct memory transfers.

Life safety systems in the areas housing this equipment have been tested in accordance with NFPA 72 and NFPA 75 requirements.

16. Memory ECC Error Injection & Correction Telemetry

Memory ECC error injection testing verified single-bit error correction and multi-bit uncorrectable error logging functionality.

Hardware ECC logic corrected 100% of injected single-bit errors with zero application performance impact.

Uncorrectable multi-bit error injection successfully triggered immediate Slurm job termination and node drain alerts.

Data acquisition systems responsible for logging these parameters employ redundant sensors to ensure measurement continuity.

17. Liquid Cooling Thermal Resistance & Pressure Drop Validation

Liquid cooling thermal resistance testing measured temperature delta between GPU silicon junction and secondary loop coolant fluid.

Thermal resistance averaged 0.042 degrees C per Watt, confirming optimal thermal interface material (TIM) application and cold plate contact pressure.

Coolant pressure drop across the chassis manifold remained within strict 4.5 PSI design limits.

Acoustic noise levels generated by the equipment in this section comply with OSHA permissible exposure limits for occupied spaces.

18. Baseboard Management Controller Redfish API Verification

Baseboard Management Controller (BMC) Redfish API performance testing evaluated telemetry streaming latency under high query loads.

Redfish API endpoints responded within 45 milliseconds to concurrent query requests from DCIM monitoring servers.

Telemetry data accuracy was verified against physical calibrated multimeter and thermal camera measurements.

Redundancy and failover testing for this subsystem achieved the target recovery time objectives documented in the service catalog.

19. Slurm Job Scheduler Dispatch & Backfill Performance

Slurm scheduler dispatch benchmarking tested job submission throughput, backfill evaluation speeds, and node state transitions.

The scheduler processed 500 job submissions per second with backfill evaluation cycles completing in under 1.5 seconds.

Database accounting logs recorded 100% accurate job execution state and resource consumption metrics.

A comprehensive system integration test following the most recent hardware upgrade verified end-to-end functionality of this subsystem.

20. Containerized Application Execution Overhead Audit

Containerized execution benchmarking evaluated Singularity/Apptainer container launch overhead and file system I/O latency.

Container launch overhead measured under 1.2 seconds compared to native host execution, with zero measurable impact on MPI compute performance.

Overlay file system I/O throughput matched bare-metal file system execution speeds.

Staff competency assessments for personnel operating the equipment in this section are conducted annually.

21. Network Packet Drop & PFC Congestion Control Testing

Network Priority Flow Control (PFC) testing verified lossless RoCEv2 transport on secondary Ethernet management networks.

Under 95% line-rate burst traffic, PFC pause frames successfully prevented buffer overrun with zero packet drops recorded.

Congestion notification (ECN) marking dynamically throttled buffer queues before pause frame activation.

Supporting documentation for the values cited in this section is filed under the annual infrastructure audit records.

22. Benchmark Test Environment Topology & Software Flags

Benchmark test environment configuration documentation records exact software versions, environment variables, compiler flags, and library paths.

System configuration manifests enable 100% reproducible benchmark execution for future acceptance testing or performance verification.

All benchmark source code, compilation scripts, and raw execution logs are archived in open-access repositories.

Data acquisition systems responsible for logging these parameters employ redundant sensors to ensure measurement continuity.

23. Benchmark Validation Variance & Reproducibility Metrics

Benchmark variance analysis across all 128 nodes confirmed node-to-node performance variance remained below 1.2% across identical workloads.

Tight performance distribution proves manufacturing consistency and uniform thermal cooling across all server racks.

Statistical outlier detection algorithms flagged zero anomalous compute nodes.

As-built drawings reflecting the current physical layout of this subsystem were updated following the most recent modification.

24. GPU Accelerator VRAM Memory Bandwidth Verification

GPU VRAM memory bandwidth benchmarking using NVML utilities verified sustained 3.32 TB/s memory transfer rates per H100 GPU.

VRAM bandwidth utilization reached 99.1% of advertised theoretical peak bandwidth across all 512 installed GPUs.

VRAM ECC overhead measured less than 0.8% total bandwidth impact.

Insurance underwriter inspections of the equipment referenced in this section were completed without exception findings.

25. System Architecture Benchmark Sign-Off Audit

OS jitter impact analysis measured operating system timer interrupt frequency and kernel background thread interference.

Kernel tuning (isolcpus, nohz_full) reduced OS jitter to under 15 microseconds per second, eliminating micro-stalls in parallel MPI ranks.

Jitter reduction yielded a 4.2% speedup in high-concurrency collective communication benchmarks.

Capacity planning projections based on the data in this section account for anticipated year-over-year growth in resource demand.

26. GPU VRAM Memory Bandwidth Verification (NVML Benchmarking)

High-frequency financial computing latency profiling evaluated sub-microsecond tick-to-trade execution paths across FPGA and GPU accelerators.

Execution latency achieved sub-500 nanosecond kernel execution times, demonstrating system suitability for ultra-low latency computational modeling.

PCIe pass-through throughput maintained zero packet drop performance under continuous tick data streams.

Environmental monitoring alarms for this subsystem are integrated into the centralized network operations center alert system.

27. System Interrupt & OS Jitter Impact Analysis

The technical specifications and measured performance characteristics for System Interrupt are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Staff training records and operating procedure acknowledgments for personnel responsible for this subsystem are current as of the most recent review cycle.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.

28. High-Frequency Financial Computing Latency Profiling

System configuration records for High-Frequency Financial Computing Latency Profiling are maintained under version control, with each modification tracked through the institutional change management process.

Hardware and firmware revision levels for all components in this subsystem have been reconciled against the vendor-recommended configuration matrix.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

29. PCIe Gen5 Peer-to-Peer DMA Throughput Testing

The technical specifications and measured performance characteristics for PCIe Gen5 Peer-to-Peer DMA Throughput Testing are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

30. Baseboard Management Controller Redfish Telemetry Audit

System configuration records for Baseboard Management Controller Redfish Telemetry Audit are maintained under version control, with each modification tracked through the institutional change management process.

Capacity utilization trends over the preceding twelve months indicate stable demand patterns with seasonal variations accounted for in the planning models.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.

31. System Interrupt & OS Jitter Benchmark Analysis

Infrastructure parameters associated with System Interrupt have been validated through a combination of automated monitoring and manual inspection procedures.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Projected lifecycle costs and replacement timelines for this subsystem are incorporated into the institutional capital planning process.

Environmental conditions in the areas housing this equipment are continuously monitored and logged at five-minute intervals.

32. High-Frequency Financial Computing Latency Profiling

The technical specifications and measured performance characteristics for High-Frequency Financial Computing Latency Profiling are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

33. Benchmark Validation Variance & Reproducibility Metrics

The technical specifications and measured performance characteristics for Benchmark Validation Variance are documented in the subsections below, with all values reflecting the most recent quarterly assessment period.

Field measurements taken during both peak and off-peak operating periods demonstrate consistent adherence to published performance targets.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.

34. GPU Accelerator VRAM Memory Bandwidth Verification

System configuration records for GPU Accelerator VRAM Memory Bandwidth Verification are maintained under version control, with each modification tracked through the institutional change management process.

Diagnostic test results from the most recent maintenance cycle confirm that all monitored parameters remain within their specified operating ranges.

Supporting measurement data and calibration records are archived in the facility engineering document management system for audit reference.

Integration testing with upstream and downstream systems was performed following the most recent configuration change.

35. System Architecture Benchmark Sign-Off Audit

System configuration records for System Architecture Benchmark Sign-Off Audit are maintained under version control, with each modification tracked through the institutional change management process.

Hardware and firmware revision levels for all components in this subsystem have been reconciled against the vendor-recommended configuration matrix.

Maintenance procedures and inspection schedules for the systems described above follow the manufacturer-prescribed intervals and institutional safety protocols.

Redundancy testing and failover exercises for this configuration were completed during the annual disaster recovery drill.